AI101 MINI-PROJECT | IIT ROPAR

github.com/jujharsingh1532/Cricket-score-predictor

Cricket Score Prediction Using Linear Regression

Predicting final innings total from live match situation parameters

PROJECT TEAM & RESPONSIBILITIES

Souvik Das
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Problem Definition & Dataset

`ipl_score_prediction.csv`

The Objective

Forecast final match totals using dynamic live match indicators to assist tactical decision-making during innings progression.

Practical Impact

Provides dynamic par scores for team captains, target setting, and real-time broadcaster analytics.

DATASET DIMENSIONS

214 Rows × 6 Columns

FEATURE VARIABLES (X)

Overs, Current Runs, Wickets

TARGET VARIABLE (Y)

Final Runs

SAMPLE RECORD TRAJECTORY

`Match ID 1: 5.1 Overs | 42 Runs | 1 Wicket → Final Score: 168`

Methodology & Pipeline

STEP 01
Data Ingestion
Load cleaned IPL match CSV records into Pandas DataFrame.

STEP 02
Match Split
GroupShuffleSplit by Match_ID (80% Train, 20% Test) avoiding data leakage.

STEP 03
Model Fitting
Train LinearRegression estimator on live parameters (Overs, Runs, Wickets).

STEP 04
Evaluation
Compute R^2, MAE, RMSE metrics on holdout test evaluation split.

MATHEMATICAL FORMULATION

$$Final_Runs = \beta_0 + \beta_1 \cdot Overs + \beta_2 \cdot Current_Runs + \beta_3 \cdot Wickets$$

Ordinary Least Squares (OLS)

CORE TECHNOLOGY STACK

`pandas` `scikit-learn` `matplotlib` `numpy`

SECTION 03

Experimental Results & Visual Pipeline

CODE TO PLOT EXECUTION

src/plots.py (Core Plotting snippet) **Execution Key**

```
# 1. Fetch & Execute Notebook Context
exec(full_code, exec_globals) y_test, y_pred = exec_globals['y_test'],
exec_globals['y_pred']

# 2. Render Scatter Plot & Fit Line

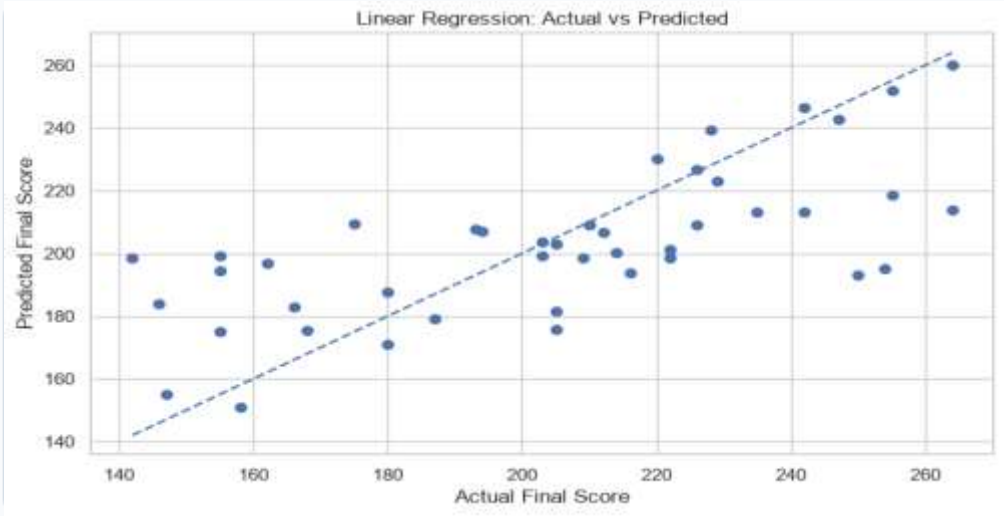
sns.scatterplot(x=y_test, y=y_pred, color='#1f77b4', s=65)
plt.plot([min_val, max_val], [min_val, max_val], 'r--')

# 3. Export Figure to Output Folder
plt.savefig('output/PredictedVSActual.png', dpi=300)
```

Full script available in repo src/main.ipynb

RENDERS

Generated Plot Output `output/PredictedVSActual.png`



R² SCORE
0.48
Explains ~55% of score variance

MEAN ABSOLUTE ERROR
19.27
Average deviation in runs

ROOT MEAN SQUARED ERROR
25.37
Penalizes larger outliers

SECTION 04

Key Learnings & Team Contributions

PROJECT RETROSPECTIVE

KEY ADVANTAGES

Linear regression provides immediate interpretability and fast inference without overfitting on small tabular match summaries.

MODEL LIMITATIONS

Higher error rates occur during early overs (0–5) due to high variance before innings trajectory stabilizes.

INDIVIDUAL CONTRIBUTIONS

Member	Role	Key Ownership
Souvik Das	Docs Lead	Data cleaning script & README structure
Jujhar Singh	Exec Lead	Model training script & GroupSplit logic
Shrinath Dixit	PPT Lead	Plotting notebook & Slide presentation
Randhi Pujitha	Video Lead	Video walkthrough & Demo recording

AI-USE DISCLOSURE

Gemini was utilized for debugging code paths, resolving Git upload errors, and formatting visual layouts.

✓ All data generation & model training executed independently.